APS360 Project Proposal: Speech-to-Text Recognition with Accents

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Abstract

This project aims to develop an artificial neural network for automatic speech recognition, with a particular emphasis on transcribing audio from non-native speakers. By leveraging deep learning techniques and a tailored neural network architecture, the system aims to enhance the accuracy and efficiency of transcribing non-native English speakers. The project addresses the need for inclusive speech recognition software that accommodates diverse accents, considering the multicultural nature of the team and the importance of effective communication across different backgrounds.

The proposal explores various models and approaches, including Hidden Markov Models (HMMs), Gaussian Mixture Models (GMMs), N-gram models, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and hybrid combinations. Deep learning and hybrid models have significantly contributed to advancements in speech processing and recognition, tackling challenges posed by foreign accents and improving overall performance. Ethical considerations, such as data sanitization, quality checks, and representative training data, are also highlighted to ensure accurate and accountable speech-to-text transcription. The ultimate goal is to develop a robust and inclusive speech recognition system that enhances communication and accessibility in academic and professional settings for individuals with diverse accents. —-Total Pages: 7

1 Introduction

The proposed project focuses on developing an artificial neural network for automatic speech recognition, with a particular emphasis on transcribing audio from non-native speakers. The motivation behind this project stems from the need for accurate and efficient speech recognition software that can accommodate diverse accents.

Traditional speech recognition models face challenges in accounting for factors such as tone of voice, gender, age, and background noise[1]. Deep learning algorithms have shown promise in addressing these challenges, but there is still room for improvement, especially for non-native speakers.

The primary objective of this project is to streamline the transcription process by leveraging a neural network specifically designed to handle non-native speech. By utilizing state-of-the-art deep learning techniques, the proposed system aims to enhance accuracy and efficiency in transcribing non-native English speakers.

An additional motivation for this project is the multicultural nature of our group. With team members from various backgrounds, the importance of developing speech recognition software that accommodates foreign accents was evident. By considering the unique characteristics and pronunciation

patterns exhibited by individuals with diverse accents, the proposed system will ensure more inclusive and effective communication.

By combining the power of deep learning algorithms, a tailored neural network architecture, and a diverse dataset, this project aims to improve automatic speech recognition, particularly for non-native speakers. The ultimate objective is to develop a robust and inclusive speech recognition system that caters to individuals with diverse accents, to enhance communication and accessibility in various academic and professional settings.

2 Illustration of Overall Model

The illustration of the RNN model for speech recognition

Figure 1: Illustration of speech recognition model

3 Background

Automatic speech recognition (ASR) is a complex field focused on converting spoken audio into written text. Developing efficient algorithms for accurate speech recognition poses challenges due to various factors such as grammar, word frequency, punctuation, tone, speed, background noise, and more [1][2].

Different procedures and algorithms are employed in speech recognition systems. Initially, a combination of Hidden Markov Models (HMMs) and Gaussian Mixture Models (GMMs) was commonly used. This approach involved analyzing audio to identify phonetic characteristics, and matching them with corresponding phrases or words using probability dependencies and statistics (e.g., grammar, word frequency) [3][4].

Another algorithm used in ASR is the N-gram model, which calculates the probability of the next word based on its relative frequency, aiding in sentence or phrase generation.

With the advent of deep learning, new ASR models incorporating architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) emerged. These architectures have shown promising results in speech recognition tasks [2]. Notably, the use of RNNs with Long Short-Term Memory (LSTM) cells and architectures like Connectionist Temporal Classification (CTC) and RNN Transducer have been explored [5].

Hybrid approaches combining CNNs and HMM-GMM models have also shown improvement in speech recognition. These hybrids utilize features such as Mel Frequency Cepstral Coefficients (MFCCs) and prediction cepstral features for preprocessing, enhancing the recognition process [6].

With the help of deep learning different models were created using a combination of new and old methods with different ways of preprocessing data as shown below [6]. The input is a speech segment that is processed using an MFCC or PLP to then be classified for an ANN to finally pass it through an HMM model approach. This model is compared with a model that directly uses the soundwave (without preprocessing) and passes it to a CNN using 2ms seconds frequency. The CNN model without preprocessing presented better results with fewer training parameters compared to conventional methods.

Another major improvement for ASR was the introduction of RNN which helped to improve the error by 16 percent (error as accents, and background noise between others) [5]. Later a new model based on an RNN was implemented by improving a model with a sound stereo input link.

Overall, the integration of deep learning and hybrid models has significantly contributed to advancements in speech processing and recognition, addressing challenges posed by foreign accents and improving overall performance.

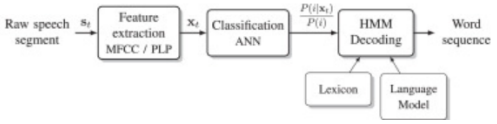

Figure 2: Fig from [6]

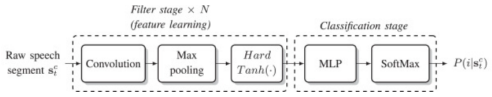

Figure 3: Fig CNN proposed [6]

4 Description of Data Sources

The datasets we will be using to train our model are called AccentDB and Common Voice, both found on Kaggle [7][8]. AccentDB includes clips of speech samples in English, spoken by both native and non-native English speakers. Both native and non-native speakers come from various countries and native languages, ensuring a broad range of accents. Common Voice is a collection of speech data gathered from the Common Voice Mozilla website, collected for the purpose of training automatic speech recognition systems. The data in this dataset is already classified by age, gender, and accent if the speaker submitted the information with the audio clip. Each sample is also voted up or down correlating to how many listeners believed the audio matches the supposed transcription of the clip.

In order to utilize these speech samples, there are several steps that may need to be taken to arrange the data into a usable format. Firstly, audio processing is crucial. This involves converting between file types to get a usable format, and ensuring a consistent sampling rate as large variations may impact the performance of the model. Additionally, removing background noise and normalizing the audio levels and pitch across the clips will improve the quality of the data. Another step that may need to be taken is manipulating the speed of the audio samples, either slowing them down or speeding them up for a more consistent talking speed across audio samples. Furthermore, it is important to label the accent of each sample to analyze its impact on the model's performance. Lastly, to ensure a well-balanced dataset, we need to consider the representation of various demographic factors. This includes equal representations of gender, accents, age, etc.

By following these steps, we can effectively clean and format our data for training a robust and inclusive speech-to-text model.

5 Description of Types of Neural Networks

We plan to use a recurrent neural network (RNN) with long short-term memory (LSTM). We have chosen to use an RNN due to its established effectiveness in speech recognition, as highlighted in a study by the University of Toronto [9]. RNNs have also been used in popular speech-to-text technologies, such as Google and Siri [10]. However, we will be utilizing a variant of RNNs called long short-term memory (LSTM) to allow the model to "selectively retain (or forget) information over longer periods of time" [10]. LSTMs accomplish this by employing specialized hidden units, which have demonstrated better effectiveness than conventional RNNs [11].

6 Description of a Simple Baseline Model

Prior to current deep learning techniques for speech recognition tasks, the conventional method used a hybrid Hidden Markov Model-Gaussian Mixture Model (HMM-GMM) machine learning approach.

The GMM was utilized to perform acoustic modeling by getting the probability distribution of acoustic features for each type of speech sound (phenome). GMMs employ a mixture of Gaussian components to represent these probability distributions, where each component represents a distinct variation of acoustic features [12].

The HMMs model how phonemes change over time and transition between different phonemes. Each phoneme is represented as a hidden state (S) and the HMM consists of a sequence of these hidden states. The transitions between the hidden states are controlled by probabilities representing how likely it is to move from one state to the other. Overall, the HMM captures the sequential nature of speech and the transitions between different phonemes [12].

During the training of the GMM-HMM method, the model learns the parameters of the GMMs and HMMs using the Baum-Welch algorithm which is a variation of the Expectation-Maximization algorithm[12]. This algorithm estimates the parameters of the HMM and GMM by iteratively updating the model based on the acoustic features and their alignment with the phonetic labels.

The last stage is to decode the probabilities to predict the likely sequence of hidden states from the acoustic features. To decode the probabilities, the Viterbi Algorithm is used[12].

7 Ethical Considerations

The difficulty of accurately transcribing speech raises ethical concerns regarding the accountability of speech-to-text models. Errors in transcription can result in miscommunication and the spreading of misinformation, prompting the question of who should be held responsible for these mistranslations and the potential harm they may inflict upon users [13].

One of the primary reasons for the low accuracy of these models is the quality of the training data. Typically, speech-to-text models are trained on large volumes of open-source data from the internet. However, if this data is not appropriately sanitized or vetted for quality, it can introduce biases toward accents, speech impediments, or other factors. Additionally, it may provide inadequate inputs for the model to learn from, further compromising its accuracy [14].

While RNNs like LSTMs significantly improved speech recognition, there are still limitations to these models. For instance, RNNs typically rely on language models like GMMs or HMMs to enhance their performance. However, language models are often biased towards standard accents and dialects, which can affect their ability to accurately recognize and transcribe speech from foreign accents. The lack of specific language models tailored to foreign accents may result in decreased performance.

To address these ethical concerns, it is crucial to enhance the accuracy and accountability of speech-to-text models. This can be achieved through rigorous data sanitization, quality checks, and the incorporation of diverse and representative training data [14]. By striving for these improvements, we can work towards ensuring fair and reliable transcription while minimizing the potential harm caused by inaccuracies.

8 Project Plan and Deadline

Code of Conduct:

- Demonstrate integrity, honesty, reliability
- Appropriate language
- Inclusivity
- Being supportive/available
- Do not let conflicts arouse hatred or contempt toward one another
- Referring to the conflict resolution process when a conflict arises.

Decision-Making Guidelines:

- Democratic majority vote after valid justification

- PEE (Point, Evidence, and Explain)
- CER (Claim, Evidence, and Reasoning)

Internal Communication:

- Discord (Formal discussions)
- WhatsApp Group Chat (discussions for clarification that only need quick responses)
- File Sharing
- When2Meet
- If a team member is tagged in a WhatsApp/Discord chat, they should try to reply in a period of 24 hours

Meeting Scheduling:

- Tuesday at 7:00 PM – 9:00 PM EST
- Sunday at 12:00 PM – 3:00 PM EST

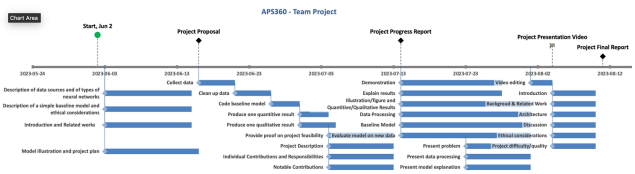

Figure 4: Breakdown of project timelines.

Tasks

Start	End	Duration (days)	Label	Assigned To
2023-06-03	2023-06-15	12	Description of data sources and of types of neural networks	Mithila Ali
2023-06-03	2023-06-15	12	Description of a simple baseline model and ethical considerations	Sagarika Rabindranath
2023-06-03	2023-06-15	12	Introduction and Related works	Cesar Vargas Santini
2023-06-03	2023-06-15	13	Model illustration and project plan	Luis Antunes
2023-06-16	2023-06-21	5	Collect data	Mithila Ali & Cesar Vargas Santini
2023-06-21	2023-06-26	5	Clean up data	Sagarika Rabindranath & Luis Antunes
2023-06-26	2023-06-30	4	Code baseline model	Mithila Ali & Cesar Vargas Santini
2023-06-30	2023-07-04	4	Produce one quantitive result	Luis Antunes
2023-06-30	2023-07-04	5	Produce one qualitative result	Sagarika Rabindranath
2023-07-04	2023-07-13	9	Provide proof on project feasibility	Mithila Ali & Cesar Vargas Santini
2023-07-04	2023-07-13	9	Project Description	Sagarika Rabindranath
2023-07-04	2023-07-13	9	Individual Contributions and Responsibilities	Luis Antunes
2023-07-04	2023-07-13	9	Notable Contributions	Mithila Ali & Cesar Vargas Santini
2023-07-23	2023-08-01	9	Present problem	Mithila Ali
2023-07-23	2023-08-01	9	Present data processing	Cesar Vargas Santini
2023-07-23	2023-08-01	9	Present model explanation	Luis Antunes & Cesar Vargas Santini
2023-07-14	2023-07-28	14	Demonstration	Sagarika Rabindranath
2023-07-14	2023-07-28	14	Explain results	Mithila Ali & Sagarika Rabindranath
2023-08-01	2023-08-04	3	Video editing	Luis Antunes
2023-08-04	2023-08-10	6	Introduction	Luis Antunes
2023-07-14	2023-08-01	18	Illustration/figure and Quantitive/Qualitative Results	Cesar Vargas Santini
2023-08-04	2023-08-10	6	Backgroud & Related Work	Sagarika Rabindranath
2023-07-14	2023-08-01	18	Data Processing	Mithila Ali & Cesar Vargas Santini
2023-08-04	2023-08-10	6	Architecture	Mithila Ali & Sagarika Rabindranath
2023-07-14	2023-08-01	18	Baseline Model	Luis Antunes
2023-07-14	2023-08-01	18	Evaluate model on new data	Sagarika Rabindranath
2023-08-04	2023-08-10	6	Discussion	Mithila Ali
2023-08-04	2023-08-10	6	Ethical considerations	Luis Antunes
2023-08-04	2023-08-10	6	Project difficulty/quality	Cesar Vargas Santini

Figure 5: Breakdown of tasks

9 Potential Risks

There are several potential risks we need to be aware of while undertaking this project; model training taking longer than expected, a team member experiencing a personal emergency, a team member dropping the course, and the dataset not being a fair representation of all accents.

To address the risk of the model training taking longer than usual, we need to investigate the cause, which can range from a small bug in the code to having a dataset that's too large. We can also improve our model's efficiency by tweaking the hyperparameters. If none of the mentioned changes

Milestones

Date	Label
2023-06-02	Start, Jun 2
2023-06-16	Project Proposal
2023-07-14	Project Progress Report
2023-08-04	Project Presentation Video
2023-08-11	Project Final Report

Figure 6: Breakdown of project milestones

speed up the training of the model, we will need to reevaluate and reprioritize deadlines to accommodate the extended training time. Throughout the process, it is crucial to document the challenges faced and learn from them for future reference.

Another risk we should consider is if a team member experiences a personal emergency. In such cases, it is important to reach out to them through various channels of communication to show support and inquire about their well-being. If their absence affects project progress, we may need to reassess deadlines and seek assistance from teaching assistants or other available resources to mitigate the impact.

Additionally, there is a risk of a team member dropping the course. In such a situation, it is crucial to evaluate which part of the project will be most impacted by their departure. By identifying their responsibilities, we can redistribute tasks among the remaining team members accordingly, ensuring that all aspects of the project are covered.

Furthermore, we need to acknowledge the risk of the dataset not being a fair representation of all accents. To mitigate this, we can consider recording our own audio clips if needed or exploring alternative datasets that provide a more comprehensive representation of different accents. By taking these measures, we can ensure a more inclusive and representative training dataset.

By proactively addressing these potential risks and implementing appropriate countermeasures, we can minimize their impact and increase the chances of a successful project outcome. Open communication, flexibility, and adaptability will be key in overcoming these challenges and ensuring the project's overall success.

10 Link to Github

https://github.com/Sagarika2870/speech-recognition

11 Citations, figures, tables, references

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